

Random Signal Theory–EEE 554

Lecture Notes 2

Discrete and Continuous Random Variables

[Reading: Kay, Sections 5.3, 5.4, 5.8, 10.3, 10.4, 10.6]

Continuous RV (CRV), Cumulative Distribution Function (CDF) [Kay, Section 10.6]

CDF $F_X(x)$ of CRV X is a continuous function (no discontinuities) if its sample space S_X has values from the real line $x \in \mathbb{R}$

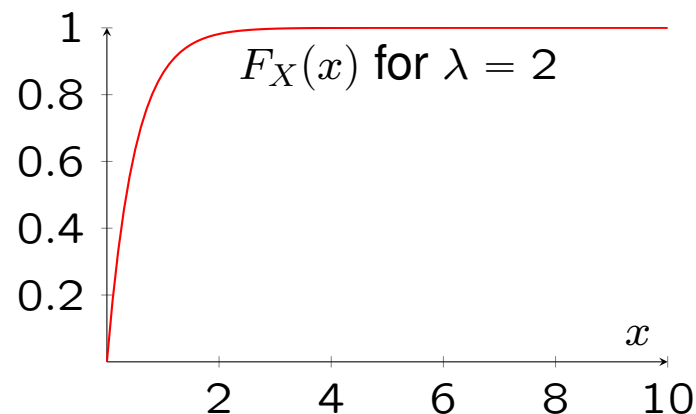
Example CRV CDF:

Given transmission time X of messages in communications system, $\lambda > 0$, satisfies

$$\Pr(X > x) = \begin{cases} 1, & x \leq 0 \\ e^{-\lambda x}, & x > 0 \end{cases}$$

then, by definition, the CDF of X is a continuous, monotonically increasing function

$$F_X(x) = \Pr(X \leq x) = \Pr(X < x) = 1 - \Pr(X > x) = \begin{cases} 0, & x \leq 0 \\ 1 - e^{-\lambda x}, & x > 0 \end{cases}$$



Properties of CDF (for both DRVs and CRVs)

P1. $F_X(\infty) = \Pr(X \leq \infty) = 1 \Rightarrow$ all sample space. Also $F_X(-\infty) = \Pr(X \leq -\infty) = 0$

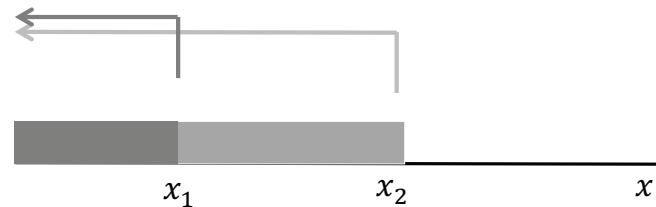
P2. If $x_1 \leq x_2 \Rightarrow F_X(x_1) \leq F_X(x_2)$
(since CDF is a monotonically increasing function)

P3. Let $F_X(x_0) = 0$; then $F_X(x) = 0$ for $\forall x \leq x_0$
(since CDF is a monotonically increasing function)

P4. $\Pr(X > x) = 1 - \Pr(X \leq x) = 1 - F_X(x)$

P5. $\Pr(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_1)$ (valid for DRV's and CRV's)

This follows since $F_X(x_2) - F_X(x_1) = \Pr(X \leq x_2) - \Pr(X \leq x_1)$



To demonstrate for DRV's: Let $x_1, x_2 \in \mathbb{N}$

- x_1 is not included in region $x_1 < X \leq x_2$
- region $X \leq x_2$ includes x_1 since $x_1 < x_2$
- region $X \leq x_1$ includes x_1
- in $\Pr(X \leq x_2) - \Pr(X \leq x_1)$, x_1 added and then removed

For continuous RV's

$$\begin{aligned}\Pr(x_1 < X \leq x_2) &= \Pr(x_1 \leq X \leq x_2) \\ &= \Pr(x_1 < X < x_2) = \Pr(x_1 \leq X < x_2) = F_X(x_2) - F_X(x_1)\end{aligned}$$

P6. For discrete RVs, the following hold

$$\Pr(x_1 \leq X \leq x_2) = F_X(x_2) - F_X(x_1) + \Pr(X = x_1)$$

$$\Pr(x_1 < X < x_2) = F_X(x_2) - F_X(x_1) - \Pr(X = x_2)$$

$$\Pr(x_1 \leq X < x_2) = F_X(x_2) - F_X(x_1) + \Pr(X = x_1) - \Pr(X = x_2)$$

Example:

DRV X equally maps N coded messages to integers $S_X = \{-2, -1, 0, 1, 2\}$ using

$$\begin{aligned} \text{if } N \geq 100, & \quad x = 2 \\ \text{if } 80 \leq N < 100, & \quad x = 1 \\ \text{if } 60 \leq N < 80, & \quad x = 0 \\ \text{if } 20 \leq N < 60, & \quad x = -1 \\ \text{if } N < 20, & \quad x = -2 \end{aligned}$$

Find $\Pr(45 < N \leq 90)$ as $\Pr(-1 \leq X \leq 1) = F_X(1) - F_X(-1) + \Pr(X = -1)$

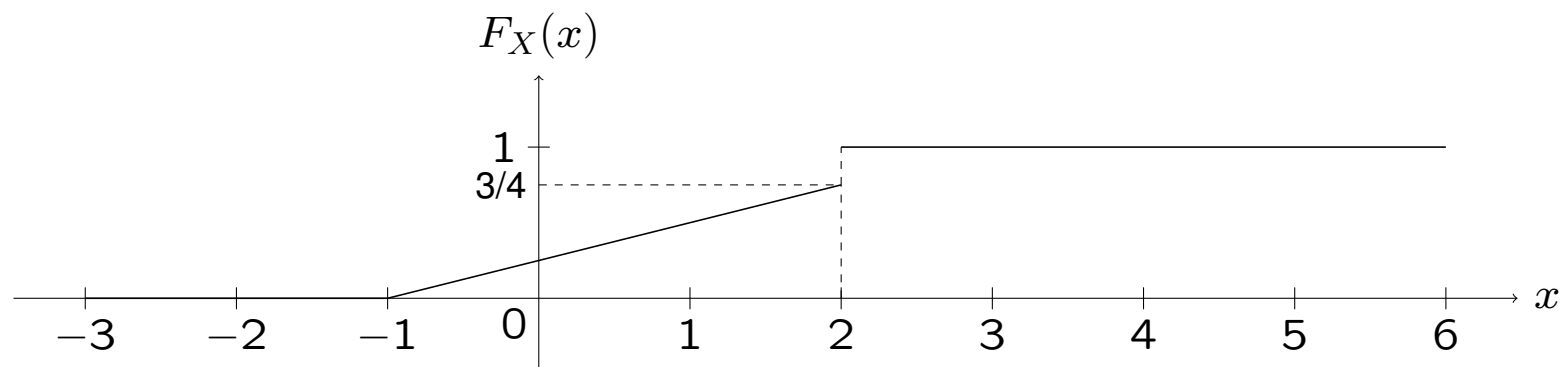
Since CDF is $F_X(x) = 0.2 u(x + 2) + 0.2 u(x + 1) + 0.2 u(x) + 0.2 u(x - 1) + 0.2 u(x - 2)$
 $F_X(1) = 0.8$, $F_X(-1) = 0.4$, and $\Pr(X = -1) = 0.2 \Rightarrow \Pr(-1 \leq X \leq 1) = 0.8 - 0.4 + 0.2 = 0.6$

Alternative approach: $\Pr(-1 \leq X \leq 1) = \Pr(X = -1) + \Pr(X = 0) + \Pr(X = 1) = 0.6$

CDF for Mixed RVs: includes a finite number of discontinuities

Example of mixed RV:

$$F_X(x) = \left(\frac{x+1}{4}\right) \left(u(x+1) - u(x-2)\right) + u(x-2) = \begin{cases} 0, & x < -1 \\ \frac{x+1}{4}, & -1 \leq x < 2 \\ 1, & x \geq 2 \end{cases}$$



RV takes a discrete value at $x = 2 \Rightarrow$ CDF has a discontinuity (jump) at $x = 2$

For mixed RV X , $\Pr(X = 2) = 1/4 \Rightarrow$ size of discontinuity at $x = 2$

A mixed RV X takes a finite set of integer values $x = x_i \in \mathbb{Z}$;
its CDF $F_X(x)$ at $x = x_i$ has a discontinuity of size $\Pr(X = x_i)$

Probability Mass Function (PMF) for discrete RV (DRV) [Kay, Section 5.4]

Interpretation: provides the likelihood of a DRV falling within a range of integer values

PMF $f_X(x)$ of DRV X with $S_X = \{x_1, \dots, x_M\}$, $x_i \in \mathbb{Z}$, is the derivative of CDF

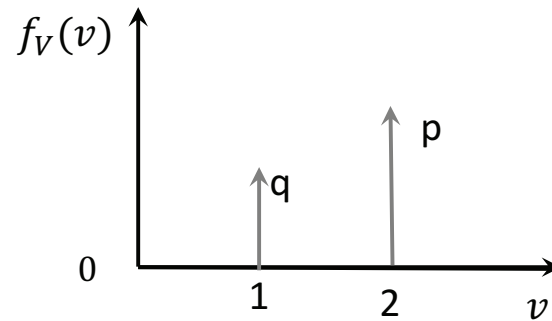
$$f_X(x) = \frac{d}{dx} F_X(x) = \frac{d}{dx} \left(\sum_{i=1}^M \Pr(X = x_i) u(x - x_i) \right) = \sum_{i=1}^M \Pr(X = x_i) \delta(x - x_i)$$

$\Rightarrow$ linear combination of delta functions since $\delta(x) = \frac{d}{dx} u(x)$

- Example of DRV voltage V of word parity bit (see page 7)

$$f_V(v) = \frac{d}{dv} F_V(v) = \frac{d}{dv} (q u(v-1) + p u(v-2)) = q \delta(v-1) + p \delta(v-2)$$

where $\delta(v - v_0) = 0$ for all v values except for $v = v_0$



- Use PMF to find probabilities:

$$\Pr(x_1 \leq X \leq x_2) = \int_{x_1^-}^{x_2^+} f_X(x) dx$$

where x^+ and x^- denote values just slightly larger and just slightly smaller, respectively, than x
this ensures that both x_1 and x_2 are included as part of integral limits

- Example of DRV X that maps N coded messages (see page 11)

Taking derivative of CDF, $f_X(x) = \sum_{n=-2}^2 \Pr(X=n) \delta(x-n)$, where $\Pr(X=n)=0.2$

$$\begin{aligned} \Pr(-1 \leq X \leq 1) &= \int_{x=-1^-}^{1^+} f_X(x) dx = \int_{x=-1^-}^{1^+} \left(\sum_{n=-2}^2 \Pr(X=n) \delta(x-n) \right) dx \\ &= \sum_{n=-2}^2 0.2 \left(\int_{x=-1^-}^{1^+} \delta(x-n) dx \right) = \sum_{n=-1}^1 0.2 \left(\int_{x=-1^-}^{1^+} \delta(x-n) dx \right) \\ &= \sum_{n=-1}^1 0.2 = 0.6 \end{aligned}$$